UNIVERSITY OF THE ARMED FORCES "ESPE"

SOFTWARE REQUIREMENTS SPECIFICATION

UniStock

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Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes UniStock, a system for managing products, sales, and inventory in a uniform and stationery store. Its organization follows the general structure of IEEE Std 830-1998 and is based on the supplied Product Backlog.

1.2 Scope

UniStock will register uniform and stationery products, classify them, store uniform sizes and schools, store stationery brands and types, register sales and calculate subtotal, taxes, and total, update stock after sales, allow customers to view and search products, provide low-stock alerts, and manage users and roles.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
Administrator	Manages products, inventory-related functions, users, and roles.
Salesperson	Registers sales.
Customer	Views and searches products.
Product	A uniform or stationery item managed by the store.
Stock	Available quantity of a product.
SRS	Software Requirements Specification.
IEEE 830	IEEE Recommended Practice for Software Requirements Specifications, used as a structural reference.
US001–US009	The nine user stories in the supplied Product Backlog.

1.4 References

- UniStock Product Backlog, Byte y Fabric Devs, University of the Armed Forces “ESPE”.
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.

1.5 Overview

Section 2 describes the system and users. Section 3 contains exactly nine functional requirements, one for each user story in the original backlog. Sections 4–6 describe quality considerations, data concepts, and acceptance checks.

2. Overall Description

2.1 Product Perspective

UniStock is a proposed application that centralizes product information, sales registration, and stock information. The Product Backlog does not specify a technology stack, database, or deployment platform.

2.2 Product Functions

- Register uniform and stationery products.
- Classify products as uniforms or stationery.
- Register sizes and schools for uniforms.
- Register stationery brands and types.
- Register sales and calculate subtotal, taxes, and total.
- Update stock after each sale.
- Allow customers to view and search products.
- Provide low-stock alerts to administrators.
- Manage system users and roles.

2.3 User Classes and Characteristics

User class	Description
Administrator	Registers/classifies products, maintains product information, receives low-stock alerts, and manages users and roles.
Salesperson	Registers sales and uses automatic subtotal, tax, and total calculation.
Customer	Views and searches products to find uniforms or stationery.

2.4 Operating Environment

The operating environment is not specified in the Product Backlog. The project team must select the platform, operating system, database, and hosting approach.

2.5 Design and Implementation Constraints

- The project is an object-oriented programming project.
- The system must support uniform and stationery product information.
- Stock must be updated after each sale.
- Access to inventory and sales functions must be controlled through user roles.
- The tax rate and low-stock threshold are not defined in the backlog.

2.6 Assumptions and Dependencies

- Administrators provide correct product information.
- The project team will define tax calculation rules and the low-stock threshold.
- Product and stock information must be available to support sales registration.
- The backlog does not specify how low-stock alerts are delivered.

3. Specific Requirements

The functional requirements below preserve the exact count of nine user stories in the supplied Product Backlog. No additional functional requirements have been introduced.

3.1 Functional Requirements

Requirement ID / Source	Name	Functional requirement
FR-001 (US001)	Product registration	The system shall allow an administrator to register uniform and stationery products so that the store's inventory can be managed.
FR-002 (US002)	Product classification	The system shall allow an administrator to classify products as uniforms or stationery so that they remain organized.
FR-003 (US003)	Uniform information	The system shall allow an administrator to register sizes and schools for uniforms so that different options can be offered to customers.
FR-004 (US004)	Stationery information	The system shall allow an administrator to register stationery brands and types so that each product can be correctly identified.
FR-005 (US005)	Sales registration and calculation	The system shall allow a salesperson to register sales and automatically calculate the subtotal, taxes, and total.
FR-006 (US006)	Inventory update	The system shall update stock after each sale so that inventory remains accurate.
FR-007 (US007)	Product viewing and search	The system shall allow customers to view and search products so that they can find the uniforms or stationery they need.
FR-008 (US008)	Low-stock alerts	The system shall provide low-stock alerts to administrators so that they can restock products.
FR-009 (US009)	User and role management	The system shall allow an administrator to manage system users and their roles so that access to inventory and sales functions can be controlled.

3.2 External Interface Requirements

3.2.1 User Interfaces

The user interface must support the nine functions described in Section 3.1. Specific screen layouts are not defined by the Product Backlog.

3.2.2 Hardware Interfaces
No specific hardware interfaces are defined in the Product Backlog.

3.2.3 Software Interfaces
No external software interfaces are defined in the Product Backlog.

3.2.4 Communications Interfaces
No communication protocols or alert-delivery channels are specified.

4. Nonfunctional Requirements
The supplied Product Backlog does not define measurable nonfunctional requirements. The following are proposed quality considerations, not additional functional requirements, and should be confirmed by the team.

4.1 Security
- NFR-SEC-001: The system should authenticate users who access administrator or salesperson functions.
- NFR-SEC-002: The system should enforce permissions associated with each user's role.
- NFR-SEC-003: If passwords are used, they should not be stored as plain text.
- NFR-SEC-004: Unauthorized users should not be able to modify inventory, sales, users, or roles.

4.2 Reliability and Data Integrity
- NFR-REL-001: Sale registration and the corresponding stock update should be consistent.
- NFR-REL-002: The system should validate input data and display understandable error messages.
- NFR-REL-003: The system should avoid duplicate stock deductions for the same completed sale.

4.3 Usability
- NFR-USA-001: Interface labels and messages should be clear to administrators and salespeople.
- NFR-USA-002: Search results should present enough information to distinguish products.

4.4 Performance and Maintainability
- NFR-PER-001: The project team should define response-time targets based on the intended environment and data volume.
- NFR-MNT-001: Product, sales, inventory, and user/role responsibilities should be organized in maintainable classes or modules.

5. Data Requirements
These data groups are implied by the backlog and do not represent additional functional requirements.

Data group	Information implied by the backlog
Product	Product identity and classification as uniform or stationery.
Uniform information	Sizes and schools associated with uniforms.
Stationery information	Brand and type.

Sale	Values needed to calculate subtotal, taxes, and total.
Inventory	Available stock quantity for products.
User and role	User identity, role, and access permissions.

6. Acceptance and Verification

Each acceptance check corresponds directly to one of the nine functional requirements.

Test ID	Acceptance check	Requirement
AT-001	Register a uniform or stationery product and verify it is recorded.	FR-001
AT-002	Classify a product as uniform or stationery and verify the classification.	FR-002
AT-003	Register sizes and schools for a uniform and verify the information.	FR-003
AT-004	Register a stationery brand and type and verify the information.	FR-004
AT-005	Register a sale and verify subtotal, taxes, and total are calculated automatically.	FR-005
AT-006	Complete a sale and verify stock is updated accordingly.	FR-006
AT-007	Search for and view a product as a customer.	FR-007
AT-008	Reach the agreed low-stock condition and verify an administrator receives an alert.	FR-008
AT-009	Manage a user's role and verify access to inventory and sales functions is controlled.	FR-009

Appendix A. Product Backlog Traceability

Backlog ID	Backlog priority	SRS functional requirement
US001	High	FR-001
US002	High	FR-002
US003	High	FR-003
US004	Medium	FR-004
US005	High	FR-005
US006	High	FR-006
US007	Medium	FR-007
US008	Low	FR-008
US009	Medium	FR-009